# Singular learning theory at finite $n$

## Effective theories and Newton polytopes

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Principles of Intelligence

Iliad Intensive 2026 · Lecture 1 of 2

# Today

- SLT at finite $n$: when it works, when it fails

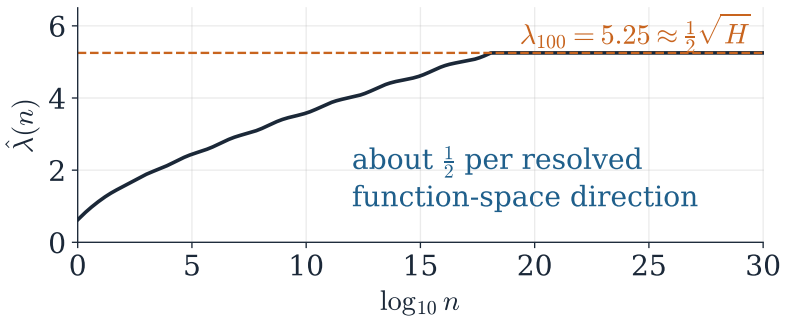
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- Along the way: effective field theory, Newton polytopes

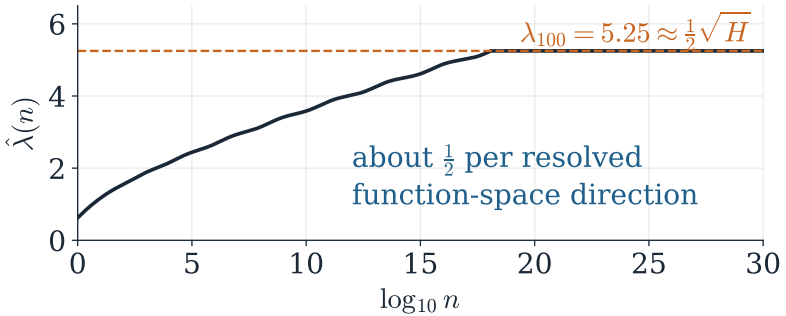
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- Along the way: effective field theory, Newton polytopes
- Next time: large width and exact RLCTs are one theory

# What we'd like to be able to say



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100 cosine neurons learning 0: about $\frac{1}{2}$ per resolved direction, then $\approx \frac{1}{2}\sqrt{H}$. (Heuristic.)

# Reminders

# Free energy

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small $F$ $\;\leftrightarrow\;$ generalizes well

# Watanabe

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So: understand $\lambda$, understand learning?

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2. Generic learning problems are not singular.
   Perturb anything and $\lambda = d/2$.

But a perturbation much smaller than $1/n$ barely changes $F(n)$.

# Effective theories

# Two scales

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smaller $n$

$\Downarrow$

larger possible $\delta K$

$\Downarrow$

possibly more singular $K_0$

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- $\hat{\lambda}(n) \to \lambda$ as $n \to \infty$
- it is the average of an *observable*

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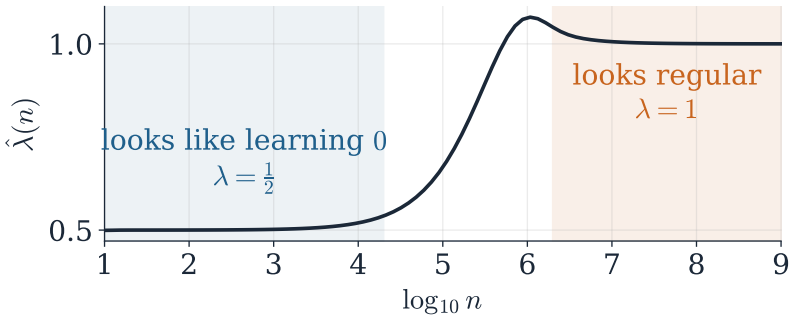
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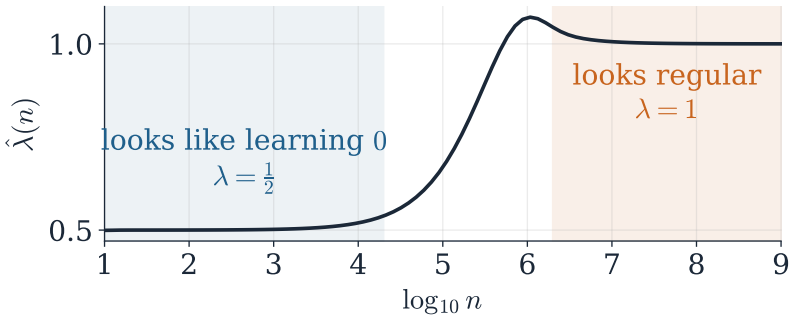
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**Hope:** at each $n$, a nearby *singular* model predicts all of them.

# Sometimes the hope is right



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One neuron, target of size $\varepsilon = 0.01$. Below $n \approx \varepsilon^{-2}$ it is learning 0; above, it is regular.

# Good news, bad news

**Good news.** Function space can be cut down to *low-degree polynomials.*

Degree about $\log n$; valid up to $n$ exponential in model size. Computations become algebraic.

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Degree about $\log n$; valid up to $n$ exponential in model size. Computations become algebraic.

**Bad news.** Some coefficients are comparable to $1/n$.

Then there is no single nearby singular model: SLT needs an extra scale.

# The simplest example

$$K(a,b) = a^2 + \varepsilon\, b^2, \qquad \varepsilon \approx n^{-1/2}$$

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Worse degeneracies: Newton polygons, in the exercises.

# The Hermite basis

# Setting

Two-layer network, Gaussian inputs $x \sim N(0,1)$:

$$f_\theta(x) = \sum_{i=1}^{H} a_i \cos(w_i x), \qquad K(\theta) = \mathbb{E}_x\big(f_\theta(x) - y^*(x)\big)^2$$

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Also works for other analytic activations, non-Gaussian bounded data, more depth.

# Hermite polynomials

$\mathrm{He}_k$: degree $k$, orthogonal for the Gaussian

$$\mathrm{He}_0 = 1, \quad \mathrm{He}_1 = x, \quad \mathrm{He}_2 = x^2 - 1, \quad \mathrm{He}_3 = x^3 - 3x, \ \ldots$$

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$c_k(\theta)$: coordinates of $f_\theta$ in this basis (normalized)

$$K(\theta) = \sum_k \left(c_k(\theta) - c_k^*\right)^2$$

# Cosine neurons

$$c_{2j}(\theta) \;=\; \pm\frac{1}{\sqrt{(2j)!}}\sum_i a_i\, e^{-w_i^2/2}\, w_i^{2j}, \qquad c_{\text{odd}} = 0$$

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- polynomial in the weights
- shrinks like $1/\sqrt{(2j)!}$

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**Theorem**

If $(2k+2)! \geq 2nH^2$, then

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For $y^* = 0$, or any polynomial target of degree $\leq 2k$. Proof: a few lines (backup, Exercise 5.2a).

# What this buys

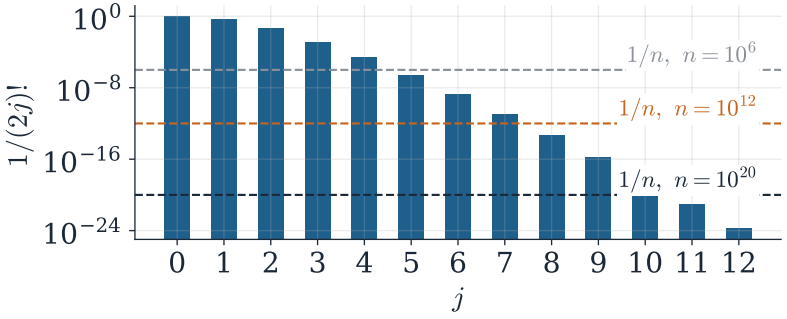
$$\text{weights} \xrightarrow{\text{polynomial}} \mathbb{R}^{k+1} \xrightarrow{\text{distance}^2} \text{loss}$$

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| $H = 100$ (200 weights) | $n = 10^6$ | $10^{12}$ | $10^{20}$ |
|---|---|---|---|
| coordinates kept | 7 | 10 | 13 |

# Which directions survive




Coordinate $j$ counts once its weight $1/(2j)!$ is above $1/n$.

# The bad news comes back

The weights $1/(2j)!$ are comparable to $1/n$

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- Exercises: an example by hand

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- Exercises: an example by hand
- Next time: this is a *feature*.
  It gives $\lambda$ at all widths at once.

# Exercises (Notebook 5)

| | | |
|---|---|---|
| 5.2a | The cosine integral and truncation | core |
| 5.3a | Newton polygons in one parameter | core |
| 5.1b | The two-plateau picture | lower priority |
| 5.2b | Other activations: the nearest pole | optional |
| 5.3b–c | Cancellation; Furman's cubic model | further |
| 5.4 | Tropical arithmetic | optional sheet |
| 5.1a | Why generic losses are non-singular | optional |
| Notebook 6 | Learning 0 at all widths | next lecture |

# Proof of the truncation theorem

Write $K = K_{\leq 2k} + R$. For cosine neurons with $|a_i|, |w_i| \leq 1$,

$$0 \leq R \leq H^2 \sum_{j>k} \frac{1}{(2j)!} \leq \frac{2H^2}{(2k+2)!} \leq \frac{1}{n}.$$

So $e^{-nR} \in [e^{-1}, 1]$:

- $Z$ changes by at most a factor $e$, so $F$ by at most 1;
- the posterior density changes by at most a factor $e$, hence so does every positive observable.

# The generating function

$$e^{tx-t^2/2} = \sum_{k\geq 0} \mathrm{He}_k(x)\, \frac{t^k}{k!}$$

Put $t = iw$, take real parts:

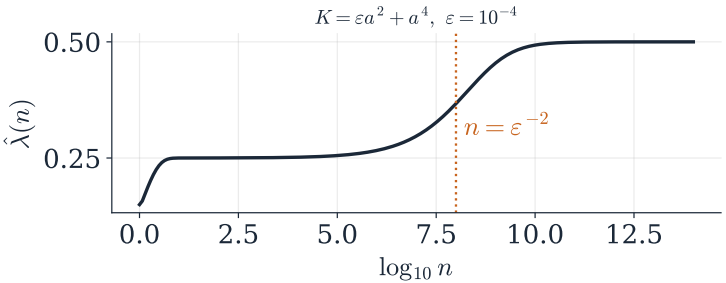
$$\cos(wx) = e^{-w^2/2} \sum_{j\geq 0} (-1)^j\, \frac{w^{2j}}{(2j)!}\, \mathrm{He}_{2j}(x)$$

# Other activations

- Analytic in a strip of half-width $\tau$: coefficients decay like $e^{-\tau\sqrt{k}}$
- tanh ($\tau = \pi/2$): about $(\log n)^2$ coordinates
- ReLU: coefficients $\sim k^{-5/4}$, about $n^{2/3}$ coordinates

Exercise 5.2b.

# Coefficients comparable to $1/n$




$K = \varepsilon a^2 + a^4$: the answer is a maximum of two lines in $\log n$. In general, a Newton polygon.